# Lesson 5: Case Study — Measles in Large and Small Towns

Source code at
pypomp/tutorials/sbied/chapter5

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## Objectives

- To display a published case study using plug-and-play methods with non-trivial model complexities.
- To show how extra-demographic stochasticity can be modeled.
- To demonstrate the use of covariates in `pypomp`.
- To demonstrate the use of profile likelihood in scientific inference.
- To discuss the interpretation of parameter estimates.
- To emphasize the potential need for extra sources of stochasticity in modeling.

## Challenges in inference from disease dynamics

Understanding, forecasting, and managing epidemiological systems increasingly depends on models. Dynamic models can be used to test causal hypotheses.

Real epidemiological systems:

- are nonlinear
- are stochastic
- are nonstationary
- evolve in continuous time
- have hidden variables
- can be measured only with (large) error

Measles is the paradigm for a nonlinear ecological system that can be well described by low-dimensional nonlinear dynamics.

A tradition of careful modeling studies have proposed and found evidence for a number of specific mechanisms, including

- a high value of $\mathcal{R}_0$ (c. 15–20)
- under-reporting
- seasonality in transmission rates associated with school terms
- response to changing birth rates
- a birth-cohort effect
- metapopulation dynamics

- fadeouts and reintroductions that scale with city size
- spatial traveling waves

Much of this evidence has been amassed from fitting models to data, using a variety of methods. See Rohani and King (2010) for a review of some of the high points.

### Measles in England and Wales

We revisit a classic measles data set, weekly case reports in 954 urban centers in England and Wales during the pre-vaccine era (1950–1963).

We examine questions regarding:

- measles extinction and recolonization
- transmission rates
- seasonality
- resupply of susceptibles

We use a model that

1. expresses our current understanding of measles dynamics
2. includes a long list of mechanisms that have been proposed and demonstrated in the literature
3. cannot be fit by previous likelihood-based methods

We examine data from large and small towns using the same model, something no existing methods have been able to do.

We ask: does our perspective on this disease change when we expect the models to explain the data in detail? What bigger lessons can we learn regarding inference for dynamical systems?

### Data sets

He et al. (2010) studied twenty towns, including

- 10 largest cities in England and Wales
- 10 smaller towns, chosen at random

Population sizes ranged from about 2,000 to 3.4 million. Weekly case reports span 1950–1963, with annual birth records and population sizes from 1944–1963.

```
all_data = UKMeasles.subset()
measles = all_data["measles"]
demog = all_data["demog"]
```

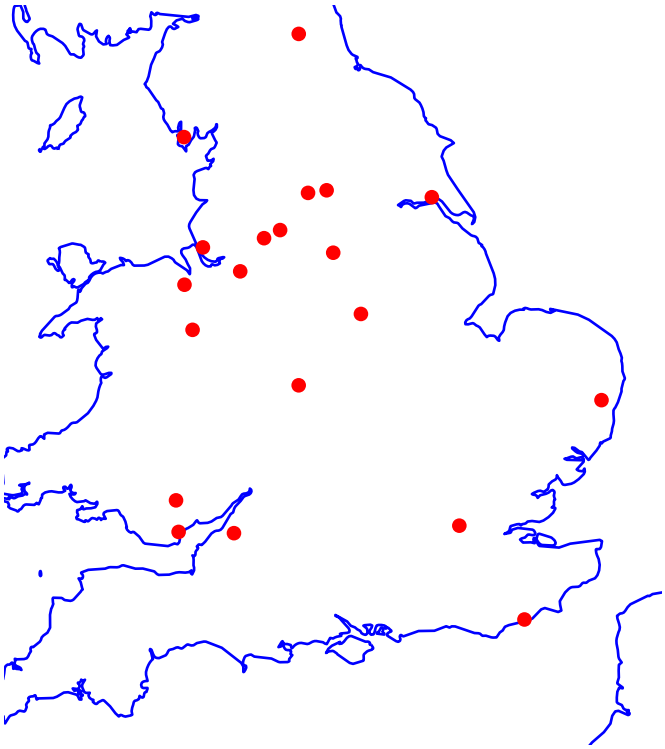
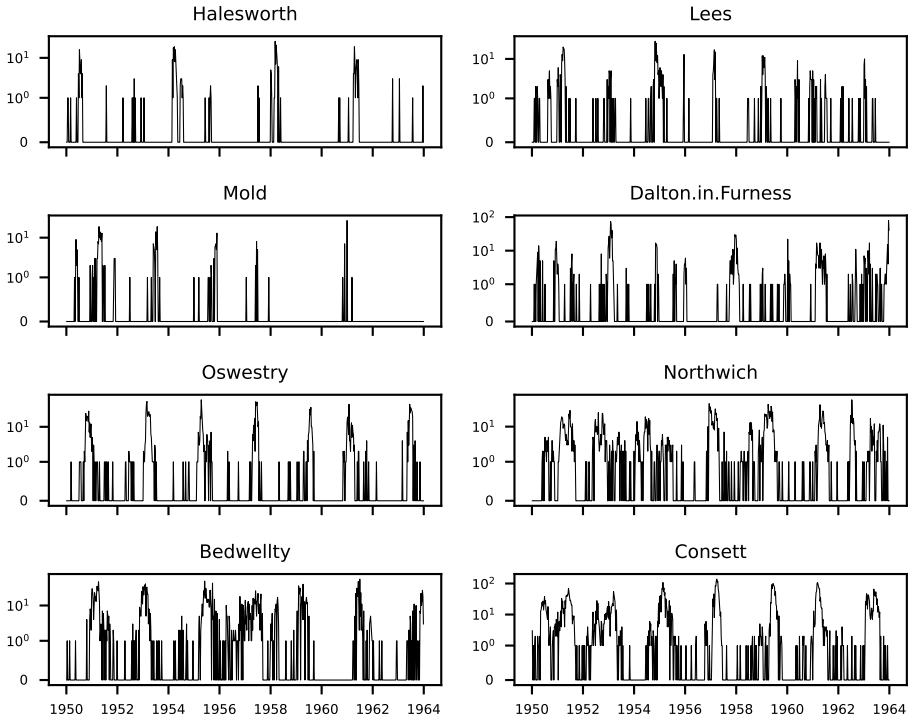


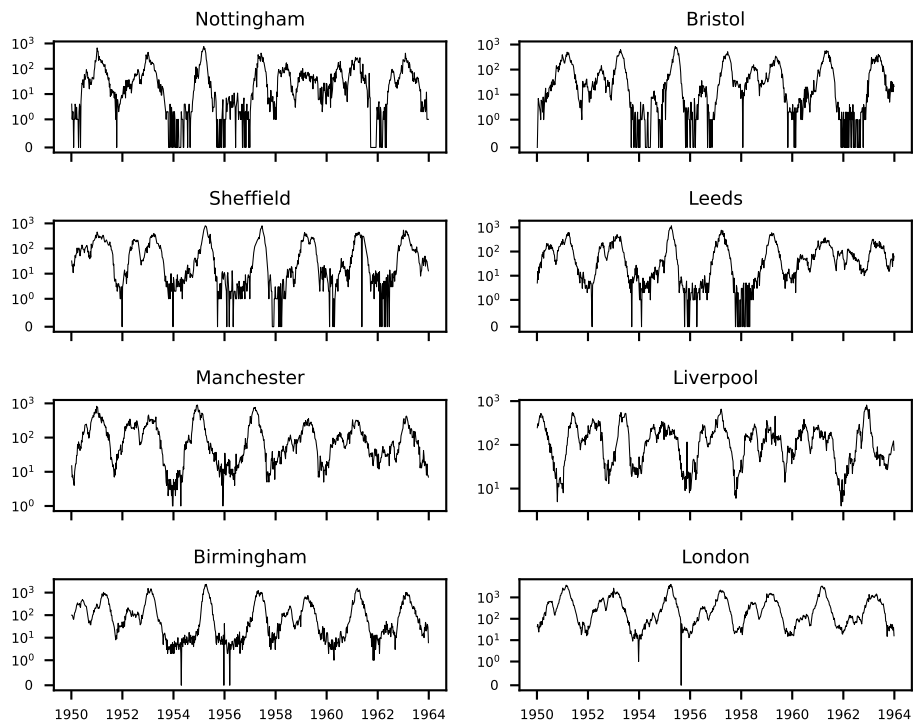
Figure 1: Locations of the 20 cities in the He et al. (2010) analysis.

## Map of cities in the analysis

## City case counts: smallest 8 cities

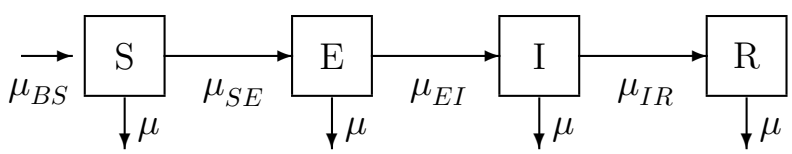


## City case counts: largest 8 cities




## Continuous-time Markov process model

The model is a SEIR (susceptible–exposed–infectious–recovered) compartment model with births and deaths:




## Model specification

**Covariates** (from data):

- $B(t)$: birth rate
- $N(t)$: population size

**Entry into susceptible class** (birth-cohort effect):

$$\mu_{BS}(t) = (1-c)\,B(t-\tau) + c\,\delta(t - \lfloor t \rfloor) \int_{t-1}^{t} B(t-\tau-s)\,ds$$

- $c$: cohort effect (fraction entering at school start)
- $\tau$: school-entry delay
- $\lfloor t \rfloor$: most recent September~1 before $t$

**Force of infection:**

$$\mu_{SE}(t) = \frac{\beta(t)}{N(t)}\,(I+\iota)\,\zeta(t)$$

- $\iota$: imported infections
- $\zeta(t)$: Gamma white noise with intensity $\sigma_{SE}$ (He et al., 2010; Bhadra et al., 2011)

**School-term transmission:**

$$\beta(t) = \begin{cases} \beta_0\,(1+a(1-p)/p) & \text{during term} \\ \beta_0\,(1-a) & \text{during vacation} \end{cases}$$

- $a$: amplitude of seasonality
- $p = 0.7589$: fraction of the year children are in school
- The factor $(1-p)/p$ ensures the average transmission rate is $\beta_0$.

**Overdispersed measurement model:**

$$\text{cases}_t \,\big|\, \Delta N_{IR} = C_t \sim \text{Normal}\big(\rho\,C_t,\ \rho\,(1-\rho)\,C_t + (\psi\,\rho\,C_t)^2\big)$$

- $\rho$: reporting rate
- $\psi$: overdispersion parameter
- When $\psi = 0$, the variance–mean relation reduces to that of the binomial distribution.

## The partially observed Markov process model

We require a simulator for our model. Notable complexities include:

1. Incorporation of the known birthrate.
2. The birth-cohort effect: a specified fraction of the cohort enters the susceptible pool all at once.
3. Seasonality in the transmission rate: higher during school terms than holidays.
4. Extra-demographic stochasticity in the form of a Gamma white-noise term acting multiplicatively on the force of infection.
5. Demographic stochasticity implemented using Euler-multinomial distributions.

## Implementation of the process model

The process model is implemented in `pypomp.measles.model_001b`. Let's walk through it.

**State variables and parameters:**

```
statenames = ["S", "E", "I", "R", "W", "C"]
accumvars = ["W", "C"]

param_names = (
    "R0", "sigma", "gamma", "iota", "rho",
    "sigmaSE", "psi", "cohort", "amplitude",
    "S_0", "E_0", "I_0", "R_0",
)
```

- $C$ is the true incidence (accumulator for new infections), reset each observation interval.
- $W$ is cumulative white noise, useful for diagnostics.

**Cohort effect:** at the start of the school year (day~251), a fraction `cohort` of the year's births enter susceptibles all at once:

```
t_mod = t - jnp.floor(t)
is_cohort_time = jnp.abs(t_mod - 251.0/365.0) < 0.5*dt
br = jnp.where(
    is_cohort_time,
    cohort * birthrate / dt + (1 - cohort) * birthrate,
    (1 - cohort) * birthrate,
)
```

**Term-time seasonality:**

```
t_days = t_mod * 365.25
in_term_time = (
    ((t_days >= 7) & (t_days <= 100))
    | ((t_days >= 115) & (t_days <= 199))
    | ((t_days >= 252) & (t_days <= 300))
    | ((t_days >= 308) & (t_days <= 356))
)
seas = jnp.where(
    in_term_time,
    1.0 + amplitude * 0.2411 / 0.7589,
    1 - amplitude
)
beta = R0 * seas * (1.0 - jnp.exp(-(gamma + mu)*dt)) / dt
```

**Extra-demographic stochasticity and transitions:**

```
# Force of infection
foi = beta * (I + iota) / pop

# Gamma white noise
dw = fast_approx_rgamma(keys[0], dt/sigmaSE**2) * sigmaSE**2

# Rates for Euler-multinomial transitions
rate = jnp.array([foi*dw/dt, mu, sigma, mu, gamma, mu])

# Poisson births
births = fast_approx_rpoisson(keys[1], br * dt)

# Euler-multinomial transitions
transitions = fast_approx_rmultinom(keys[2], populations, rt_final)
```

- `fast_approx_rgamma`, `fast_approx_rpoisson`, and `fast_approx_rmultinom` are JAX-compatible random variate generators provided by `pypomp`.

**State update:**

```
S = S + births - trans_S[0] - trans_S[1]
E = E + trans_S[0] - trans_E[0] - trans_E[1]
I = I + trans_E[0] - trans_I[0] - trans_I[1]
R = pop - S - E - I
W = W + (dw - dt) / sigmaSE
C = C + trans_I[0]
```

- Since recognized measles cases are quarantined, most infection occurs before case recognition.

True incidence $C$ counts individuals progressing from I to R.

### State initializations

The initial state allocates the population across compartments according to proportions $S_0, E_0, I_0, R_0$:

```
def rinit(theta_, key, covars, t0):
    S_0, E_0, I_0, R_0 = (
        theta_["S_0"], theta_["E_0"],
        theta_["I_0"], theta_["R_0"])
    m = covars["pop"] / (S_0 + E_0 + I_0 + R_0)
    S = jnp.round(m * S_0)
    E = jnp.round(m * E_0)
    I = jnp.round(m * I_0)
    R = jnp.round(m * R_0)
    return {"S": S, "E": E, "I": I, "R": R,
        "W": 0, "C": 0}
```

### Measurement model

We model both under-reporting and measurement error. We want $\mathbb{E}[\text{cases}|C] = \rho\, C$ and $\text{Var}[\text{cases}|C] = \rho\,(1-\rho)\,C + (\psi\,\rho\,C)^2$.

Specifically, cases $\mid C \sim f(\cdot \mid \rho, \psi, C)$, where

$$\begin{aligned} f(c \mid \rho, \psi, C) \quad = \quad & \Phi\big(c + \tfrac{1}{2},\, \rho C,\, \rho(1-\rho)C + (\psi\rho C)^2\big) \\ & -\Phi\big(c - \tfrac{1}{2},\, \rho C,\, \rho(1-\rho)C + (\psi\rho C)^2\big)\,. \end{aligned}$$

Here, $\Phi(x, \mu, \sigma^2)$ is the c.d.f. of the normal distribution with mean $\mu$ and variance $\sigma^2$.

The measurement density `dmeas`:

```
def dmeas(Y_, X_, theta_, covars, t):
    rho, psi, C = theta_["rho"], theta_["psi"], X_["C"]
    m = rho * C
    v = m * (1.0 - rho + psi**2 * m)
    sqrt_v = jnp.sqrt(v) + 1e-18
    y = Y_["cases"]
    upper = jax.scipy.stats.norm.cdf(y + 0.5, m, sqrt_v)
    lower = jax.scipy.stats.norm.cdf(y - 0.5, m, sqrt_v)
    lik = jnp.where(y > 1e-18, upper - lower, upper) + 1e-18
    return jnp.log(lik)
```

The measurement simulator `rmeas`:

```
def rmeas(X_, theta_, key, covars, t):
    rho, psi, C = theta_["rho"], theta_["psi"], X_["C"]
    m = rho * C
    v = m * (1.0 - rho + psi**2 * m)
    cases = jax.random.normal(key) * (jnp.sqrt(v) + 1e-18) + m
    return jnp.where(cases > 0.0, jnp.round(cases), 0.0)
```

### Data and covariates

The `pypomp` package includes the UK measles data from He et al. (2010). The `UKMeasles` class provides convenient access.

We illustrate using London:

```
london_data = UKMeasles.subset(units=["London"])
dat = london_data["measles"]
print(f"Date range: {dat['date'].min()}",
    f" to {dat['date'].max()}")
print(f"Number of observations: {len(dat)}")
```

```
Date range: 1944-01-07 00:00:00  to 1965-03-26 00:00:00
Number of observations: 1108
```

### The case report data

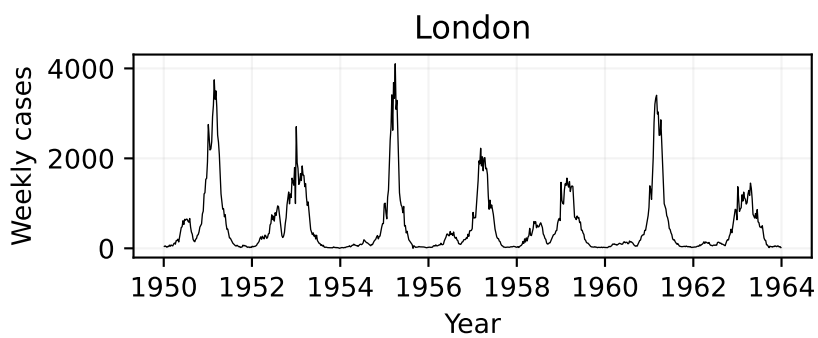



Figure 2: Weekly measles case reports, London 1950–1963.

### Reviewing covariates in time series analysis

Suppose our time series of primary interest is $y_{1:N}$. A **covariate time series** is data $z_{1:N}$ which is used to help explain $y_{1:N}$.

- It may be implicit that a covariate $z_{1:N}$ is considered an **external forcing** to the system producing $y_{1:N}$, i.e., $z_{1:N}$ causally affects $y_{1:N}$ but not vice versa.
- E.g., weather might affect human health, but human health has negligible effect on weather: weather is an external forcing to human health.

When the process leading to $z_{1:N}$ is not external to the system generating it, we must be alert to the possibility of **reverse causation** and **confounding variables**.

- Here, births are external, assuming negligible effect of measles on live births.
- For a mechanistic POMP, a covariate does not have to enter the equations linearly, as in a standard linear model.

### Covariates in Pypomp

- A covariate table is incorporated into the POMP object by the `covars` argument to `pp.Pomp`.

- The named coviariates are delivered to the model components, after look up at the corresponding time, `t`.

### Other common POMP covariates

A linear or nonlinear trend through time in any parameter or rate or latent state can be modeled using covariates.

- Seasonality can be modeled as a covariate which can enter the model in any scientifically plausible way.
- Periodic B-splines provide a flexible way to incorporate seasonality; the source code for `pp.dacca` provides an example.
- The measles model describes seasonality directly via the `t` variable accessible in all the model components.

### Covariate pre-processing for measles

- Discretely measured covariates must be interpolated at arbitrary times for a continuous-time POMP.

We interpolate the covariates and delay the entry of newborns into the susceptible pool. This is handled internally by `UKMeasles.Pomp()`, which uses smoothing splines for population interpolation and shifts the birth rate by 4 years (the school-entry delay $\tau$).

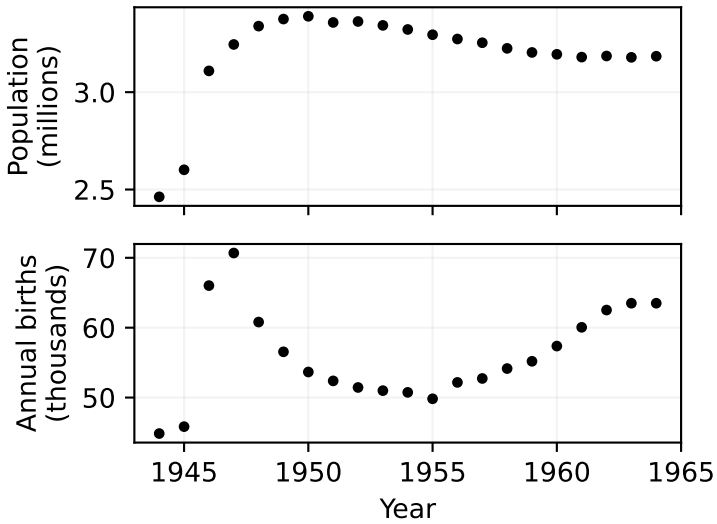



Figure 3: Population and birth-rate covariates for London.

### Constructing the POMP object

```
theta_london = mles["London"].to_dict()
```

```
m1 = UKMeasles.Pomp(
    unit=["London"],
    theta=theta_london,
    model="001b",
    interp_method="shifted_splines",
    first_year=1950,
    last_year=1963,
    dt=1/365.25,
    clean=True
)
```

```
State names: ['S', 'E', 'I', 'R', 'W', 'C']
Time range: 1950.01 to 1963.98
Observations: 730
```

### Estimates from He et al. (2010)

He et al. (2010) estimated the parameters of this model for all 20 cities. We verify that we get the same likelihood.

```
key = jax.random.key(998468235)
cache_file = cache_dir + "/london_pf.pkl"
if os.path.exists(cache_file):
    with open(cache_file, 'rb') as f:
        london_pf = pickle.load(f)
else:
    m1.pfilter(J=[50,500,5000][RL],
        reps=[2,5,10][RL], key=key, thresh=0,
        CLL=True, ESS=True)
    london_pf = m1.results_history[-1]
    with open(cache_file, 'wb') as f:
        pickle.dump(london_pf,f)

london_logliks = london_pf.logLiks.values.flatten()
ll = pp.logmeanexp(london_logliks)
ll_se = pp.logmeanexp_se(london_logliks)
print(f"Log-likelihood: {ll:.1f} (SE {ll_se:.2f})")
```

```
Log-likelihood: -3806.1 (SE 0.25)
```

### Simulations at the MLE

```
key = jax.random.key(42)
X_sims, Y_sims = m1.simulate(key=key, nsim=3)
```

### Parameter transformations

The parameters are constrained to be positive, and some lie between 0 and 1. We turn the likelihood maximization problem into an unconstrained one by transforming:

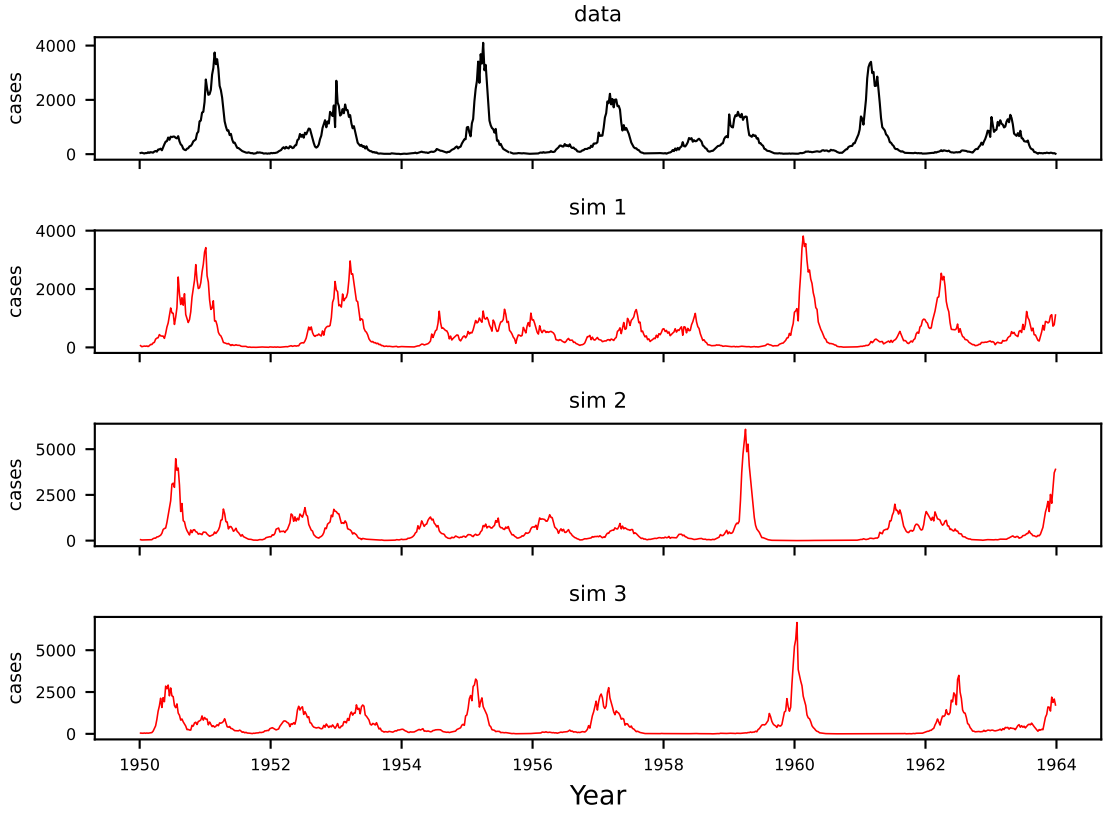



Figure 4: Simulations (red) versus data (black) at the London MLE.

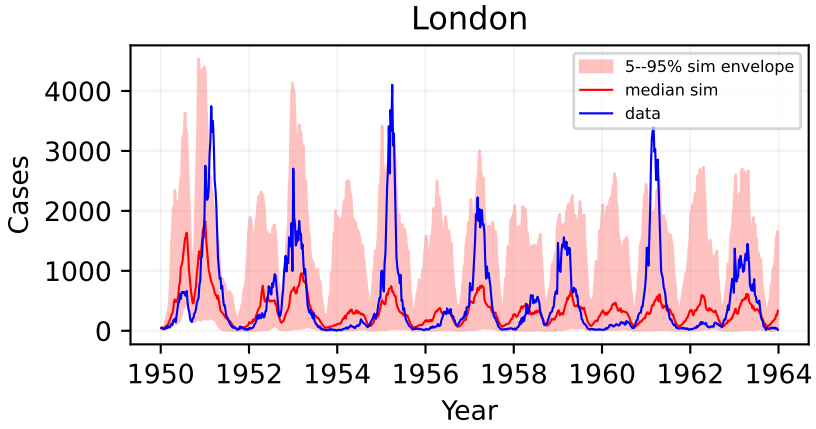



Figure 5: Simulation envelope (100 sims) versus data.

- **Positive parameters** ($R_0$, $\sigma$, $\gamma$, $\iota$, $\sigma_{SE}$, $\psi$): log transform
- **Parameters in** $(0,1)$ ($\rho$, cohort, amplitude): logit transform
- **Initial proportions** $(S_0, E_0, I_0, R_0)$: log-barycentric transform

These are implemented in `model_001b.to_est` and `model_001b.from_est`.

### ARMA benchmark

Linear, Gaussian auto-regressive moving-average (ARMA) models provide a flexible non-mechanistic benchmark.

```
from statsmodels.tsa.arima.model import ARIMA

cases = m1.ys["cases"].values
log_y = np.log(np.maximum(cases, 1))

arma_fit = ARIMA(log_y, order=(2, 0, 2)).fit()
arma_loglik = arma_fit.llf - np.sum(log_y)

print(f"ARMA(2,2) log-lik: {arma_loglik:.1f} (p=5)")
print(f"SEIR model log-lik: {ll:.1f} (p=12)")
```

```
ARMA(2,2) log-lik: nan (p=5)
SEIR model log-lik: -3806.1 (p=12)
```

- Minimizing AIC, $2p - 2\ell$, is equivalent to maximizing $\ell - p$.
- The aim of mechanistic modeling is not to beat benchmarks, but falling far behind can diagnose problems.
- "Far" means many log units: differences of log-likelihoods are invariant to the scale of measurement.

### Log-likelihood anomalies

The benchmark and model log-likelihoods can both be decomposed as a sum of conditional log-likelihoods,

$$\ell(\theta) = \sum_{n=1}^{N} \ell_n(\theta) \quad \text{where} \quad \ell_n(\theta) = \log f_{Y_n|Y_{1:n-1}}(y_n^* \,|\, y_{1:n-1}^*;\, \theta).$$

The **anomaly** for the model at time $t_n$ is the difference between the model conditional log-likelihood and that of the benchmark.

Anomalies can be used similarly to regression residuals: they can indicate points where the model fails; patterns can reveal scope for model improvement.

### Particle filter variance

**Effective sample size** is $\text{ESS} = \left(\sum_{j=1}^{J} w_j^2\right)^{-1}$, where $w_j$ are the resampling probabilities, i.e., the normalized filter weights.

- ESS is the equivalent number of independent particles.

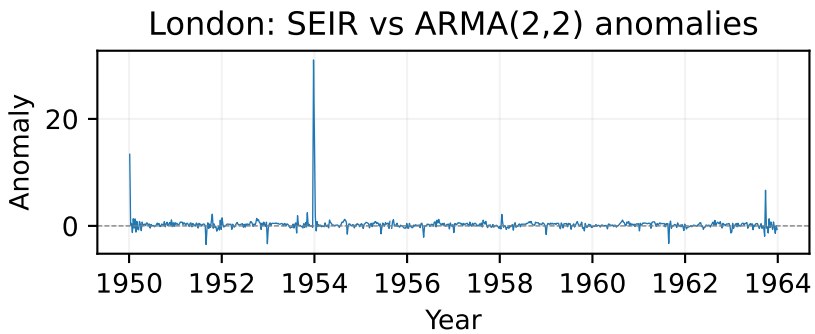



Figure 6: Conditional log-likelihood anomalies for London. Positive anomalies indicate the model outperforms ARMA; negative anomalies indicate the model struggles.

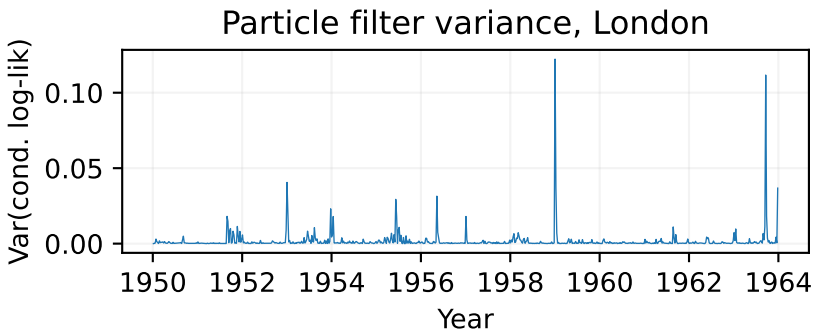



Figure 7: Variance of conditional log-likelihoods across replicates. Observations with high variance are numerically problematic.

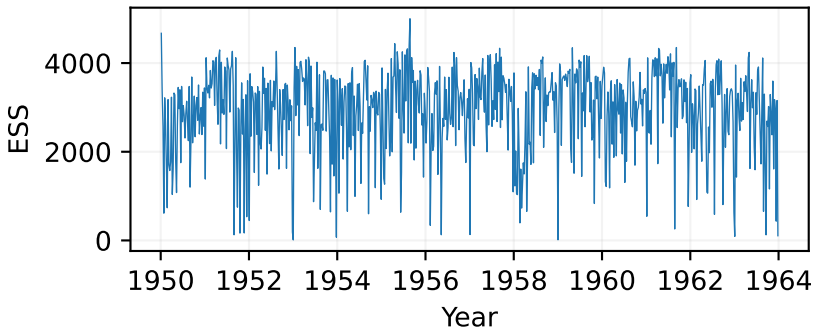



Figure 8: Effective sample size for one particle filter in London.

- If $w_j = 1/J$ for all $j$ then ESS $= J$.

### Interpretation of ESS

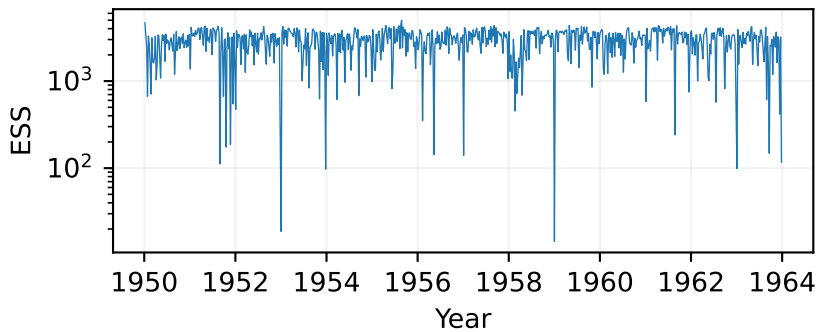



Figure 9: Mean of replicated ESS.

- As a rule of thumb, ESS $< 100$ is not good.
- Some time points here are problematic and could usefully be investigated further. Is there a problem with either the model or the data at these points?

### Results from He et al. (2010)

The fitting procedure used is as follows:

- A large number of searches were started at points across the parameter space.
- Iterated filtering was used to maximize the likelihood.
- Point estimates of all parameters were obtained for 20 cities.
- Profile likelihoods were constructed to quantify uncertainty in London and Hastings.

The linked document `lesson5_profile.qmd` shows how a likelihood profile can be constructed using IF2 in `pypomp`.

### Setting up IF2

The IF2 algorithm requires specifying random walk standard deviations for each parameter being estimated:

```
from pypomp import RWSigma

rw_sd = RWSigma(
    sigmas={
        "R0": 0.02, "sigma": 0.02, "gamma": 0.02,
        "iota": 0.0, "rho": 0.0,
        "sigmaSE": 0.02, "psi": 0.02,
        "cohort": 0.02, "amplitude": 0.02,
        "S_0": 0.02, "E_0": 0.02, "I_0": 0.02, "R_0": 0.02
    },
    init_names=["S_0", "E_0", "I_0", "R_0"]
)
```

```
m1.mif(J=2000, M=50, a=0.95, rw_sd=rw_sd,
       key=jax.random.key(1234567), thresh=0)
```

| town | pop | R0 | a | LP | IP | iota | rho | psi | sigSE |
|---|---|---|---|---|---|---|---|---|---|
| Halesworth | 2171 | 33.1 | 0.38 | 7.9 | 2.3 | 0.01 | 0.754 | 0.64 | 0.075 |
| Lees | 4247 | 29.7 | 0.15 | 8.5 | 2.1 | 0.03 | 0.612 | 0.68 | 0.080 |
| Mold | 6409 | 21.4 | 0.27 | 5.9 | 1.8 | 0.01 | 0.131 | 2.87 | 0.054 |
| Dalton.in.Furness | 10560 | 28.3 | 0.20 | 5.5 | 2.0 | 0.04 | 0.455 | 0.82 | 0.078 |
| Oswestry | 10970 | 52.9 | 0.34 | 10.3 | 2.7 | 0.03 | 0.631 | 0.48 | 0.070 |
| Northwich | 18330 | 30.1 | 0.42 | 8.5 | 3.0 | 0.06 | 0.795 | 0.40 | 0.086 |
| Bedwellty | 28930 | 24.7 | 0.16 | 6.8 | 3.0 | 0.04 | 0.311 | 0.95 | 0.061 |
| Consett | 39130 | 35.9 | 0.20 | 9.1 | 2.7 | 0.07 | 0.650 | 0.41 | 0.071 |
| Hastings | 65690 | 34.2 | 0.30 | 7.0 | 5.4 | 0.19 | 0.695 | 0.40 | 0.096 |
| Cardiff | 244600 | 34.4 | 0.22 | 9.9 | 3.1 | 0.14 | 0.602 | 0.27 | 0.054 |
| Bradford | 294300 | 32.1 | 0.24 | 8.5 | 3.4 | 0.24 | 0.599 | 0.19 | 0.045 |
| Hull | 302100 | 38.9 | 0.22 | 9.2 | 5.5 | 0.14 | 0.582 | 0.26 | 0.064 |
| Nottingham | 307000 | 22.6 | 0.16 | 5.7 | 3.7 | 0.17 | 0.609 | 0.26 | 0.038 |
| Bristol | 442600 | 26.8 | 0.20 | 6.2 | 4.9 | 0.44 | 0.626 | 0.20 | 0.039 |
| Leeds | 509700 | 47.8 | 0.27 | 9.5 | 10.9 | 1.25 | 0.666 | 0.17 | 0.078 |
| Sheffield | 515000 | 33.1 | 0.31 | 7.2 | 6.4 | 0.85 | 0.649 | 0.18 | 0.043 |
| Manchester | 704500 | 32.9 | 0.29 | 11.1 | 6.9 | 0.59 | 0.550 | 0.16 | 0.055 |
| Liverpool | 802300 | 48.1 | 0.30 | 7.9 | 9.8 | 0.26 | 0.494 | 0.14 | 0.053 |
| Birmingham | 1117900 | 43.4 | 0.43 | 8.5 | 11.6 | 0.34 | 0.544 | 0.18 | 0.061 |
| London | 3389620 | 56.8 | 0.55 | 13.1 | 12.5 | 2.90 | 0.488 | 0.12 | 0.088 |

### Some parameter estimates vary with city size

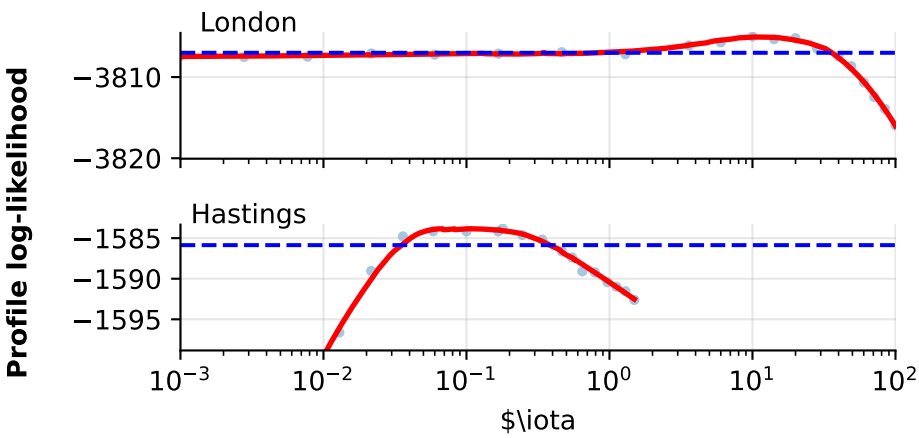
The Spearman rank correlations between each parameter and $N_{1950}$ reveal which parameters vary systematically with city size.

```
Spearman r(parameter, N_1950):
  R0          : +0.46
  a           : +0.31
  LP          : +0.31
  IP          : +0.95
  iota        : +0.93
  rho         : -0.20
  psi         : -0.93
  sigSE       : -0.31
```

### Imported infections

For implementation, see the exercise notes and their source code.

$$\text{force of infection} = \mu_{SE} = \frac{\beta(t)}{N(t)}\,(I+\iota)\,\zeta(t)$$




### Seasonality

Profile likelihoods over the amplitude of term-time seasonality show that:

- Both London and Hastings have high seasonality amplitude ($a > 0.4$).
- School terms drive measles transmission strongly.
- The confidence intervals are fairly tight, indicating this parameter is well-identified.

### Cohort effect

Profile likelihoods over the cohort entry fraction show that:

- The cohort effect is moderately well-identified in London.
- In Hastings, the profile is flatter, reflecting weaker identifiability in small populations.

### $R_0$ estimates inconsistent with literature

Recall that $\mathcal{R}_0$ is a measure of how communicable an infection is. Existing estimates ($\mathcal{R}_0 \approx 15$–$20$) come from serology surveys and feature-based methods.

- The MLE estimates of $\mathcal{R}_0$ from He et al. (2010) are much higher: $\mathcal{R}_0 \approx 40$–$60$.
- This reflects the fact that the susceptible fraction is much smaller than in a "virgin population" assumed by classical estimates.
- Profile likelihoods over $\mathcal{R}_0$ confirm this finding for both London and Hastings.

### Extrademographic stochasticity

$$\mu_{SE} = \frac{\beta(t)}{N(t)} (I + \iota) \zeta(t)$$

- $\sigma_{SE} > 0$ is strongly supported by the data.
- Without extra-demographic stochasticity, the model fits poorly.
- Profile likelihoods over $\sigma_{SE}$ also reveal trade-offs with the infectious period (IP) and latent period (LP): as $\sigma_{SE}$ increases, IP tends to decrease.

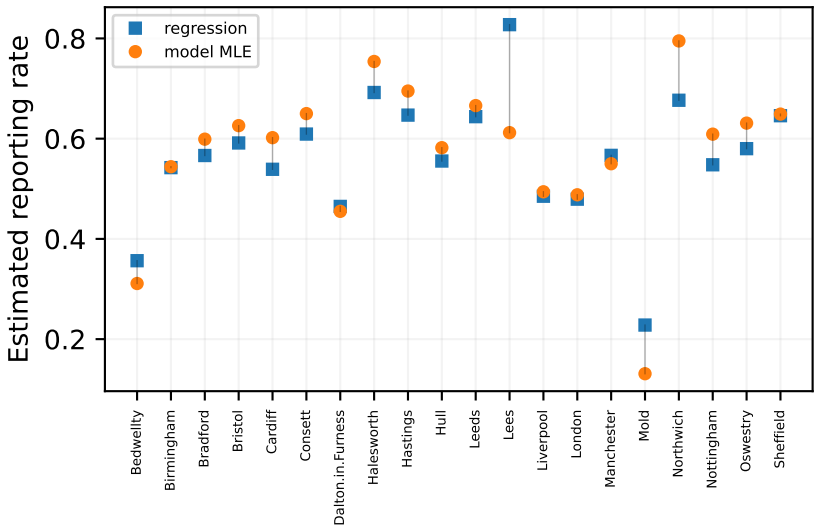



Figure 10: Reporting rate: regression estimate (cases/births slope) versus model MLE.

### Reporting rate

### Predicted vs observed critical community size

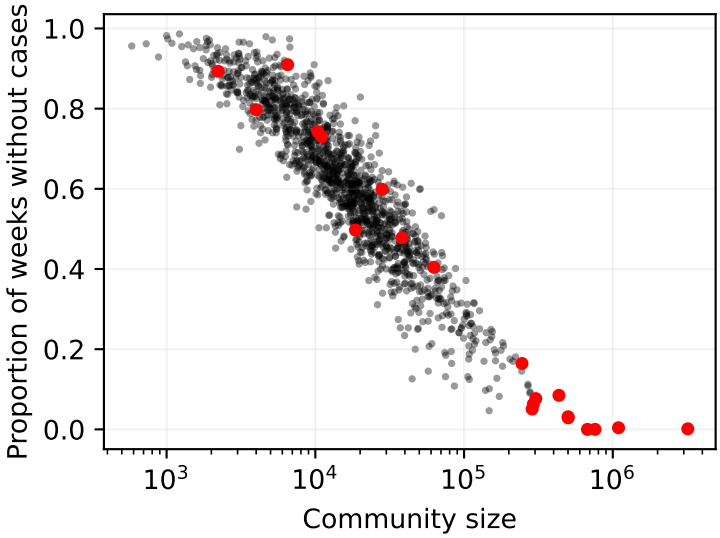



Figure 11: Fraction of weeks with zero cases versus population size. The critical community size is approximately 300,000.

### Questions

- What does it mean that parameter estimates from the fitting disagree with estimates from other data?
- How can one interpret the correlation between infectious period and city size in the parameter estimates?

- How do we interpret the need for extrademographic stochasticity in this model?

### Exercise 5.1. Reformulate the model

- Modify the He et al. (2010) model to remove the cohort effect. Run simulations and compute likelihoods to convince yourself that the resulting codes agree with the original ones for `cohort = 0`.
- Now modify the transmission seasonality to use a sinusoidal form. How many parameters must you use? Fixing the other parameters at their MLE values, compute and visualize a profile likelihood over these parameters.

### Exercise 5.2. Extrademographic stochasticity

Set the extrademographic stochasticity parameter $\sigma_{SE} = 0$, set $\alpha = 1$ (already the case in `model_001b`), and fix $\rho$ and $\iota$ at their MLE values, then maximize the likelihood over the remaining parameters.

- How do your results compare with those at the MLE? Compare likelihoods but also use simulations to diagnose differences between the models.

## References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